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The Cardiac Arrest Sonographic Assessment (CASA) exam – A standardized approach to the use of ultrasound in PEA☆

Keywords:
Cardiopulmonary resuscitation
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1. Background

Emergency physicians (EPs) have started integrating point-of-care-ultrasound (POCUS) into the evaluation of patients with cardiac arrest to identify reversible causes of pulseless electrical activity (PEA) and it is now recommended by the AHA [1-6]. However, POCUS prolongs CPR pauses which negatively impacts survival [7-10]. While POCUS protocols for cardiac arrest have been suggested, they are too complex and cumbersome [2-4,11-12]. Here we present an evidence based, three step protocol that is simple, rapid, and allows the EP to quickly and accurately identify reversible causes of PEA while minimizing CPR interruptions.

2. The CASA exam

The Cardiac Arrest Sonographic Assessment (CASA) exam consists of three, <10 second POCUS exams to occur at pulse checks (Fig. 1). We recommend the recorder time each pause out loud. A phased array (cardiac) transducer should be used and all images should be recorded for review. The EP should utilize the optimal cardiac window based on the patient, with the subxiphoid view often being preferable as this view can be performed during ongoing CPR. While the CASA exam is designed for a single provider, a second provider should perform the ultrasound when possible, as this has been associated with shorter CPR interruptions [8]. FAST and tension pneumothorax exams can be performed on a case-by-case basis during CPR.

2.1. Step one: cardiac tamponade

Pericardial effusion causing cardiac tamponade is the cause of cardiac arrest in 4–15% of patients [5,13-16]. Tamponade can be rapidly intervened upon to resolve PEA and patients with tamponade have higher survival to hospital discharge rates (15.4%) than other causes of PEA (1.3%) [5]. The examiner should assess for pericardial effusion and, if present, assess for signs of tamponade such as early diastolic right ventricular collapse (Fig. 2). If cardiac tamponade is present, a pericardiocentesis should be strongly considered.

2.2. Step two: pulmonary embolus

Pulmonary embolism (PE) is the cause of cardiac arrest in 4.0–7.6% patients and these patients have higher survival to hospital discharge rates (6.7%) than other patients in PEA (1.3%) [5,17-20]. Thus, it is important that EPs quickly and accurately assess for PE in all cardiac arrest patients. To assess for the presence of PE, the examiner should evaluate for signs of right heart strain such as a dilated right ventricle and a comparatively small left ventricle (Fig. 3). If signs of right heart strain are present, PE should remain a consideration as the cause of PEA.

2.3. Step three: cardiac activity

The presence or absence of cardiac activity on bedside echo provides useful prognostic information for patients in PEA [5]. The absence of cardiac activity in PEA portends worse outcomes. Patients in PEA with cardiac standstill on ultrasound have survival to hospital discharge rates ranging from 0.0% to 0.6% [5,21-22]. While low, these rates are not zero and thus initial resuscitation should be attempted in all patients regardless of cardiac activity. The examiner should assess for global cardiac activity or fibrillations (Supplemental video 1). If cardiac activity is present, the examiner should reassess for pulses and blood pressure and consider starting continuous vasopressors. If no cardiac activity is seen, the utility of ongoing resuscitation should be reviewed in combination with other poor prognostic factors, such as low end-tidal CO2, prolonged downtime, and unwitnessed arrest.

2.4. Ancillary steps: tension pneumothorax and FAST

Tension pneumothorax is an exceedingly rare cause of non-traumatic cardiac arrest and can often be diagnosed clinically [18]. During ongoing CPR, the EP should examine the bilateral anterior chest for the absence of lung sliding indicating pneumothorax (Fig. 4). If a pneumothorax is detected, needle decompression or thoracostomy may be considered. Small, clinically insignificant pneumothoraces can occur from rib fractures during CPR and clinicians should be aware that these injuries may not need acute intervention.

Also during ongoing CPR, the EP, if the clinical scenario indicates, can assess for a ruptured abdominal aortic aneurysm (AAA) or ectopic pregnancy by performing a FAST exam looking for evidence of free fluid [23].

Evaluation of the IVC and hypovolemia was intentionally excluded from the CASA exam as intravenous fluids are traditionally given empirically in cardiac arrest and we expect the IVC to be distended in most cardiac arrest as there is severely limited forward flow.

3. Discussion

POCUS is a powerful tool for assessing reversible causes of cardiac arrest. However, it must be utilized in a protocolized, efficient manner so as to not do harm. The CASA exam assesses for the highest yield reversible causes of PEA that can be visualized with ultrasound, while limiting

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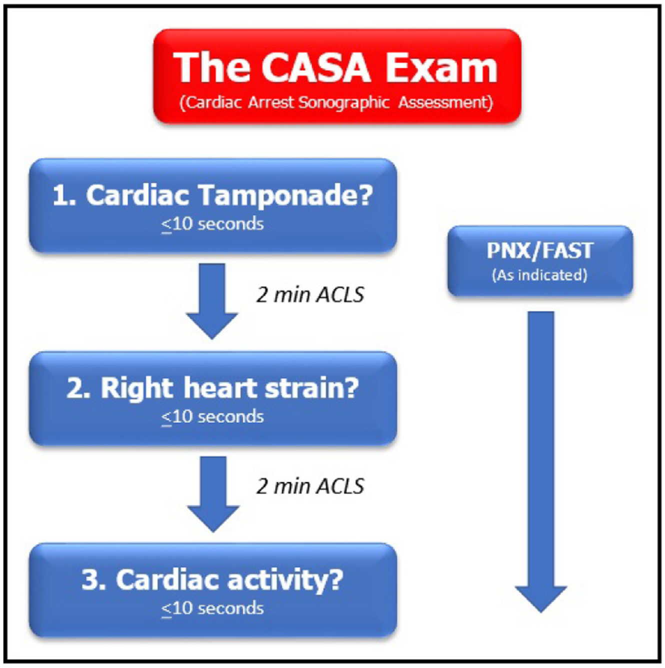

Fig. 1. The Cardiac Arrest Sonographic Assessment (CASA) exam schematic.

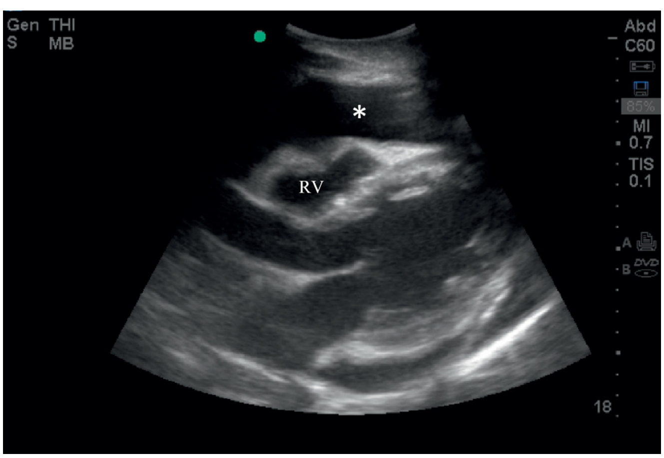

Fig. 2. Parasternal long view demonstrating pericardial effusion (*) causing cardiac tamponade. Note the early diastolic right ventricular collapse (RV).

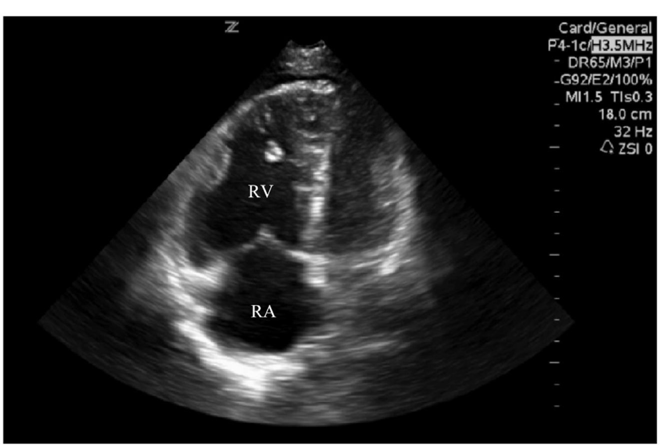

Fig. 3. Apical four chamber view demonstrating right heart strain consistent with the presence of PE. Note the dilated right atrium (RA) and right ventricle (RV).

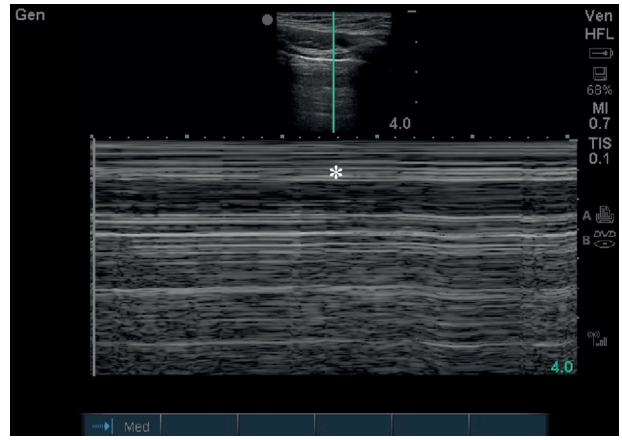

Fig. 4. US image of pneumothorax. Note the characteristic absence of lung sliding and the bar code sign in M mode (*).

POCUS's negative impact. It is our hope that the utilization of the CASA exam will limit prolonged CPR pauses and improve diagnostic accuracy.

Supplementary data to this article can be found online at http://dx.doi.org/10.1016/j.ajem.2017.08.052.

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